PRESSURIZED Final Documentation

Project Overview

PRESSURIZED is a tactile arcade game experience developed by Holden Denyer as a Senior Capstone project in the Immersive Media Design program at the University of Maryland. The project pairs a hand-built arcade cabinet with a custom physical controller- an expansive panel of real-world switches, dials, buttons, knobs, and levers - that is wired to a Unity-built submarine game. The player pilots a submarine, cruising on the ocean floor, responding to on-screen command prompts by interacting with the correct physical input on the control deck.

Project Description

PRESSURIZED is a complete arcade unit consisting of three integrated components: a Unity-developed video game, a hand-fabricated physical controller, and a hand-built plywood arcade cabinet. The game is submarine-themed and challenges players to react to real-time command prompts by locating and operating the correct physical input on the control deck. Inputs include faucet valves, a hand wheel, a stapler, sliders, toggle switches, arcade buttons, a number keypad, a telephone handset, and more- each labeled with absurd module names laser-etched directly onto the acrylic control panel surface.

Audience Experience

Upon approaching the arcade cabinet, the player sees a large glowing marquee displaying the PRESSURIZED logo in red, backlit lettering. Below it, an alarm siren light signals game state. The player steps up to the cabinet, inserts into the game via a start screen featuring a dramatic logo reveal animation, and is guided through a brief tutorial. Once gameplay begins, the on-screen game view shows an underwater scene scrolling through multiple randomized biomes. Command prompts appear in a console area, instructing the player to interact with specific modules on the physical controller. As the player earns points, boss encounters trigger: aquatic creatures appear on screen in one of multiple screen zones, and the player must press the corresponding button in a directional '+' to land hits. LED backlighting on the controller reacts to gameplay events. At the end of each life, a new element of vessel damage is revealed. By the end of the game, the window is cracked, steam valves are burst, and an alarm light blares both in game and in front of the player. The cabinet accommodates one or more players side by side, enabling a unique cooperative experience.

Visual and Audio Design

The game art is rendered in a doodle-like, hand-drawn style with distinct outlines and saturated color. The underwater world features multiple randomized biomes, each with unique environment assets and a corresponding boss creature. Audio includes ambient underwater sound, alarm effects, and reactive sounds to cue player action. The controller panel is laser-etched acrylic over a plywood base, with LED strip backlighting that responds dynamically to game state. The arcade cabinet is constructed from plywood sheets and 2x4 framing, following ADA-informed height guidelines with controls seated below 36 inches and a non-reflective matte display screen.

Showcase Information

Presented: Friday, May 8, 2026

Location: Iribe Center Main Lobby, University of Maryland

Creator: Holden Denyer

Program: Immersive Media Design, Computing Track